

NeuroSymbolic AI Assignment 2025/2026: Ordered List Reasoning using **NeurASP**

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Problem Description

In this assignment you will work with a fundamental reasoning task: given a list of five integers, determine the *sorted position* of each element.

Formally, for an input list:

$$[x_1, x_2, x_3, x_4, x_5]$$

the system must output:

$$[p_1, p_2, p_3, p_4, p_5]$$

where each p_i is an integer in the domain $\{0, 1, 2, 3, 4\}$ representing the position of x_i in the ascending ordering of the list (0 = smallest, 4 = largest).

Objective

Students must design two different systems capable of predicting the positional ranking of integers:

1. A NeuroSymbolic model based on **NeurASP**, where ASP rules define ordering constraints and guide the learning process.
2. A purely neural baseline trained with classical supervised methods (e.g. Cross Entropy Loss).

The goal is to compare the two paradigms and analyze how symbolic constraints contribute to accuracy, generalization, and training efficiency.

Assignment Tasks

1. **Symbolic Component (ASP Rules)** Define the Answer Set Programming rules needed to reason about integer ordering. Your NeurASP program should:
 - Represent pairwise order relations between the 5 integers.
 - Use ASP rules to derive consistent positions in $\{0, 1, 2, 3, 4\}$.
 - Enforce that all output positions are distinct.
 - Integrate with neural predicates declared using NeurASP's `nn` interface.
2. **Neural Component** Build a neural network that takes 5 integer as input and outputs a distribution over the 5 possible positions for each integer.
3. **NeuroSymbolic Component** Combine the neural model with ASP rules using NeurASP. The neural predicate suggestions must be consistent with:
 - integer ordering rules
 - distinctness of positions
4. **Experiments Part** Conduct a full experimental analysis:
 - Train the NeurASP model using your symbolic constraints.
 - Extract the neural network from the NeurASP model for performing evaluation
 - Train the purely neural baseline using supervised learning.
 - Evaluate both systems on:
 - Exact match accuracy
 - Per-position accuracy
 - Precision / Recall / F1
 - **Training time** (mandatory)
 - Plot learning curves.
 - Provide an analysis discussing:

- Accuracy differences between the two systems
- Differences in convergence speed
- Typical failure cases

Additional suggestions:

- Compare performance under different training set sizes (100, 1k, 10k samples).
- Analyze how symbolic rules help under low-data regimes.

5. **Theory Part** Prepare a short slide deck (8–12 slides) explaining:

- Core ideas of NeurASP.
- Interaction between neural components and ASP constraints.
- Your system architecture.
- Experimental results.
- A critical reflection on strengths and limitations.

Submission

You have two submission slots:

- Last lecture of the course: **19 December**
- First exam session (appello): **16 January**